

On the optimal design of global funds rewarding countries for pricing carbon emissions

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Abstract

Consider a global institution with an exogenous budget that can reward each country based on its tax rate on the combustion of a given fossil fuel. I develop a model in which countries differ in the co-benefits that they derive from emissions reductions and also in the intrinsic disvalue that they derive from taxing carbon (interpreted as arising due to domestic opposition). Assuming that all countries have linear demand functions for the fossil fuel, I compute the optimal mechanism for the case where the global institution can only condition its reward payments on each country's carbon tax. The optimal mechanism can be implemented through a reward payment function of the following form: each country is rewarded based on how much (if at all) its carbon tax exceeds a certain reference level. The reward payment is quadratic in this amount if the global institution's budget is small. For large budgets, there is an additional term that is linear in the amount that the carbon tax exceeds the reference level.

1 Introduction

Currently, global environmental institutions spend around \$6 billion per year on contracting with developing countries on reducing carbon emissions. This is done predominantly through project-based approaches, where the global institutions co-finance emissions-abating projects that it assesses would otherwise not get done. The current paper argues that if the world were to raise substantial additional funding for incentivising emissions abatement in developing countries then it might be optimal spent this additional money via a new kind of fund that would reward countries for pricing carbon emissions.

These carbon pricing reward funds that I propose here would each year pay monetary transfers to developing countries depending only on their net taxes (i.e. subtracting subsidies) on fossil fuels. I propose a fully operational procedure for determining these reward payment functions (see section 4) and argue in its favor relative to some possible alternative design choices (see sections 4 and 5).

The procedure and its evaluative discussion are based on a simple model. In the model, there is a global institution with an exogenous budget that can contract with countries on their taxes on a particular fossil fuel (e.g. coal, oil or more specifically “oil used for transportation”, “oil used for heating”, etc.). There is a continuum of countries choosing their net tax on the fossil fuel (where fossil fuel subsidies are viewed as negative taxes).

The global institution can only condition the reward payment to a given country on that country’s current carbon tax. I justify this assumption in section 5.1 by arguing that conditioning on other variables (such as the country’s past carbon tax) would set a bad precedent in effectively rewarding countries for globally detrimental past actions (such as past carbon taxes).

Each country’s utility is quasilinear in money and has three further parts. Firstly, there is the purely economic benefit derived from the use of the fossil fuel. This part is assumed to be public information, given that changes in the economic benefit from fossil fuel use can actually be inferred from how fossil fuel use reacts to price changes. The actual fossil fuel used in the country is determined by the tax rate on the fossil fuel, as all economic actors take the tax rate as given and optimise accordingly.

Secondly, there is a privately known local co-benefit from emissions reductions. This is interpreted to include the countries subjective valuation of the benefits of health improvements from reduced local air pollution.

Thirdly, there is a privately known intrinsic cost from taxing the fossil fuel, interpreted to arise from the popular opposition to it and the resulting societal disruptions.

I specialise to the case where the purely economic benefits from fossil fuel use are a second order polynomial of the fossil fuel use and proportional in all countries, the local co-benefits are linear in fossil fuel use and the intrinsic costs from fossil fuel taxation are linear in the tax rate.

In this case, the two privately known parameters can actually be reduced to a single linear combination which can be viewed as the type of the country. This type also equals the tax rate that the country chooses in the absence of the global institution. I assume that the types are uniformly distributed over a finite interval.

The global institution’s objective is the sum of the countries’ current welfare and the emissions reductions times the social cost of carbon. The global institution chooses an anonymous reward payment function conditioning only on a country’s choice of the net tax rate on the fossil fuel. Countries are always free to ignore the mechanism.¹

I show that the optimal mechanism can be implemented through a reward payment function of the following form: each country is rewarded based on how much (if at all) its tax on the fossil fuel exceeds a certain reference level. The reward payment is quadratic in this amount if the global institution’s budget is

¹By the revelation principle, this problem amounts to choosing an assignment of a tax rate to each type under the constraint that the participation constraints and incentive compatibility constraints are satisfied.

small. For large budgets, there is also a term that is linear in the amount that the tax exceeds the reference level.

I calibrate the model for the case of coal and oil and assuming that the global institution only rewards developing countries (specifically, non-Annex 1 countries). For this I use middle of the road estimates of the price elasticity of demand for these fossil fuels. I calibrate the distribution of types so as to reproduce the observed taxes and subsidies on coal and oil in non-annex 1 countries as well as their status quo use of coal and oil. I argue that it is appropriate to use for the social cost of carbon a value based on a normative approach, since the question is how to best design the mechanism. Based on this, I use a social cost of carbon on \$417 based on Ricke et al. (2018).

Numerically, I find that the optimal mechanism under this social cost of carbon of ends up being almost identical to the optimal mechanism in the case where the global institution only cares about reducing emissions (see section 3.4). For this case I find explicit formulae (see Lemma 8) for the parameters of the optimal reward payment function, depending only on the available budget and the parameters (elasticities, range of types). I argue that the simplicity of these formulae makes them attractive for practical application. Based on them, I propose an operational procedure for determining a reward function in practice for use if and when carbon pricing rewards are established (section 4).

2 The model

There is a continuum of countries, indexed by their type, $(\alpha, \beta) \in R^2$, explained further below. Each country chooses a domestic carbon tax rate τ . All the actors in the country's economy then face this carbon price and optimise accordingly. This determines the aggregate fossil fuel use $y(\tau)$ in the economy, as I will now detail.

$B(y)$ denotes the market surplus from fossil fuel use. As a result of facing the carbon price τ in addition to the exogenous² world market price p of fossil fuels, the actors in the economy end up jointly maximizing the following:

$$B(y) - py - \tau y$$

Let us denote by $y(\tau)$ the resulting fossil fuel use. It is characterized through:

$$B'(y(\tau)) = p + \tau$$

For tractability, I now make the following:

Assumption 1. *The benefit from using fossil fuels is given by:*

$$B(y) = \frac{y(2e_d - y + 2)}{2e_d}$$

²This exogeneity assumption is not restrictive. In fact the current model can be embedded in a global Walrasian equilibrium model developed in Stern (2021). The analysis below can be viewed as finding how the global institution can achieve a given reduction in fossil fuel demand at a minimal budget.

Lemma 1. *Fossil fuel demand is given by $y = 1 + e_D(1 - p - \tau)$.*

We thus have $y(\tau) = 1 + e_D(1 - p - \tau)$. I will later choose the normalisation that $p = 1$. It follows that e_d is the price elasticity of demand of fossil fuels in the absence of any carbon tax. I assume that this has the same value in all countries.

The country's utility is given by:

$$u(\tau, \alpha, \beta) = B(y(\tau)) - py(\tau) - \alpha y - \beta \tau$$

α can be interpreted to capture the country's valuation of local externalities from fossil fuel use. β can be interpreted to capture the costs due to popular opposition to carbon taxes and the resulting disruptions.

From now on we will reason about a country's utility relative to what it gets if it chooses its tax rate to maximise its utility (not including transfers from the global institution that I will introduce below):

Definition 1.

$$\tilde{u}(\tau, \alpha, \beta) := u(\tau, \alpha, \beta) - \sup_{\tau'} u(\tau', \alpha, \beta)$$

Lemma 2. *Let $\gamma := \alpha - \frac{\beta}{e_d}$. We have:*

$$\tilde{u}(\tau, \alpha, \beta) = -\frac{1}{2}e_d(\tau - \gamma)^2$$

We can thus view γ as the country's type. We denote by $v(\tau, \gamma)$ the change in utility that the type γ experiences when choosing the tax rate τ instead of choosing the tax rate that maximises u . We thus write:

$$v(\tau, \gamma) := -\frac{1}{2}e_d(\tau - \gamma)^2$$

Since we are assuming that e_d is identical across countries, $\gamma := \alpha - \frac{\beta}{e_d}$ is a characteristic of the country that captures all that is relevant for the country's utility. Formally, once we know γ , it does not matter through which combination of (α, β) is arises. We will therefore from now on view γ as being the country's type.

Now I introduce the global institution. This global institution has an exogenous budget M . It can observe the carbon price τ and offer countries nonnegative transfers, $t(\tau)$. The country's overall utility is thus:

$$v(\tau, \gamma) + t(\tau)$$

The global institution knows the function $B(y)$ and in particular the parameter e_d (which equals the price elasticity of demand of fossil fuel in the absence of any carbon tax, given that we will normalize the world market price p of fossil fuels to 1.). The motivation for this assumption is that in practice observable market responses to changes in fossil fuel prices can be used to infer the shape of $B(y)$.

I assume that the global institution cannot condition its transfer on γ . I discuss and motivate this assumption in section 5.1. The global institution's objective is:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} (v(\tau(\gamma), \gamma) - \eta y(\tau(\gamma))) f(\gamma) d\gamma$$

where $f(\gamma)$ denotes the density of the type distribution and η the social cost of carbon, as evaluated by the global institution.

The global institution's budget constraints means that its aggregate transfers cannot exceed its exogenous budget M :

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} t(\tau(\gamma)) f(\gamma) d\gamma \leq M$$

2.1 The optimal contract under complete information

Under complete information, the global institution's problem is to choose a carbon tax rate $\tau(\gamma)$ for each type γ to solve the following:

Problem 1.

$$\max_{\tau(\cdot)} \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma) (v(\tau(\gamma), \gamma) - \eta y(\tau(\gamma))) d\gamma$$

under the following 2 constraints:

1. exogenous budget:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma) t(\tau(\gamma), \gamma) d\gamma \leq M$$

where M denotes the global institution's exogenous budget.

2. voluntary participation:

$$v(\tau(\gamma), \gamma) + t(\tau(\gamma)) \geq 0$$

Lemma 3. *Under complete information, the optimal contract implements the following profile of tax rates:*

For $M \leq \eta^2 \frac{e_d}{2}$:

$$\tau(\gamma) = \gamma + \sqrt{\frac{2M}{e_d}}$$

The associated welfare gains w are given by

$$w = \eta \sqrt{2M e_d} - M$$

The resulting emissions reductions are given by:

$$\sqrt{2e_d M}$$

For $M \geq \eta^2 \frac{e_d}{2}$:

$$\tau(\gamma) = \gamma + \eta$$

The associated welfare gains w are given by :

$$w = \eta^2 \frac{e_d}{2}$$

The resulting emissions reductions are given by:

$$e_d \eta$$

Proof. See appendix A.1. □

2.2 The global institution's problem under incomplete information

Under incomplete information (or in the setting where the global institution can only condition on the tax rate τ but not the type τ , the following incentive compatibility constraint arises:

$$v(\tau(\gamma), \gamma) + t(\tau) \geq v(\tau', \gamma) + t(\tau') \forall \gamma, \tau'$$

Definition 2. Let us define $U(\gamma)$ as the utility gain that type γ can realise by participating relative to no participating (in which case his optimal tax rate is $\tau = \gamma$). We thus have:

$$U(\gamma) := \sup_{\tau} v(\tau, \gamma) + t(\tau)$$

By the envelope theorem, we have

$$U'(\gamma) = \frac{\partial v}{\partial \gamma} \big|_{(\tau(\gamma), \gamma)}$$

where $\tau(\gamma) := \operatorname{argmax}_{\tau} v(\tau, \gamma) + t(\tau)$.

Using Lemma 2, we get:

$$U'(\gamma) = e_d(\tau(\gamma) - \gamma)$$

It is clear that it is never optimal for the global institution to implement a τ with $\tau(\gamma) < \tau$ for any γ . Thus we have that $U(\gamma)$ is non-decreasing. Hence there exists γ_L such that all $\gamma < \gamma_L$ do not participate whilst all $\gamma > \gamma_L$ do participate. Moreover, the participation constraint must be binding at γ_L . Thus we obtain:

$$U(\gamma) = 0 \forall \gamma \leq \gamma_L$$

$$U(\gamma) = e_d \int_{\gamma=\gamma_L}^{\gamma} (\tau(\gamma) - \gamma) d\gamma \forall \gamma \geq \gamma_L$$

We also have:

$$U(\gamma) = u(\tau(\gamma), \gamma) + t(\gamma)$$

which yields:

$$t(\gamma) = -u(\tau(\gamma), \gamma) + e_d \int_{\gamma=\gamma_L}^{\gamma} (\tau(\gamma) - \gamma) d\gamma$$

Thus the global institution's problem reduces to the following:

Problem 2.

$$\max_{\tau(\cdot)} \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(v(\tau(\gamma), \gamma) - \eta y(\tau(\gamma))) d\gamma$$

under the following constraint:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} (-u(\tau(\gamma), \gamma) + U(\gamma)) d\gamma \leq M$$

where M denotes the global institution's exogenous budget and $U(\gamma)$ is given by:

$$U(\gamma) = 0 \forall \gamma \leq \gamma_L$$

$$U(\gamma) = e_d \int_{\gamma=\gamma_L}^{\gamma} (\tau(\gamma) - \gamma) d\gamma \forall \gamma \geq \gamma_L$$

Lemma 4. *Let λ denote the Lagrange multiplier associated with the budget constraint. The optimal direct mechanism satisfies for all participating γ :*

$$\eta - \lambda \frac{1 - F(\gamma)}{f(\gamma)} - (1 + \lambda)(\tau - \gamma) = 0$$

Proof. See appendix 2.1. □

From now on I will make the following:

Assumption 2. γ is distributed uniformly on $[\underline{\gamma}, \bar{\gamma}]$.

I will now define $\mu := \frac{1}{\lambda+1}$. Thus $\mu = 0$ corresponds to the case where the budget M is 0 and $\mu = 1$ to the case where the budget is sufficient to implement the first best.

Lemma 5. *For any budget $M \geq 0$, let $\mu(M) := \frac{1}{\lambda(M)+1}$, where $\lambda(M)$ denotes the global institution's shadow value of funds when its budget is M . The optimal mechanism implements the following profile of tax rates:*

$$\tau(\gamma) = \max(\gamma, \mu(M)\eta - (1 - \mu(M))(\bar{\gamma} - \gamma) + \gamma)$$

Moreover, the following holds:

$$\mu(0) = 0$$

$$\mu(M) = 1 \forall M \geq \sup\{M : M \text{ is sufficient to implement the first best}\}$$

$\mu(M)$ is strictly increasing on $[0, \sup\{M : M \text{ is sufficient to implement the first best}\}]$

For $M \in [0, \frac{e_d \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}]$ the $\mu(M)$ is given as the unique $\mu \in [0, \frac{\bar{\gamma} - \underline{\gamma}}{\bar{\gamma} - \underline{\gamma} + \eta}]$

satisfying:

$$\frac{e_d \eta^3 (2 - \mu) \mu^3}{6(1 - \mu)^2 (\bar{\gamma} - \underline{\gamma})} = M$$

and the set of participating types is given by $[\bar{\gamma} - \eta \frac{\mu}{1 - \mu}, \bar{\gamma}]$.

For $M \in [\frac{e_d \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}, \frac{1}{2} e_d \eta (\eta + \bar{\gamma} - \underline{\gamma})]$ the $\mu(M)$ is given as the unique

$\mu \in [0, \frac{\bar{\gamma} - \underline{\gamma}}{\bar{\gamma} - \underline{\gamma} + \eta}]$ satisfying:

$$\frac{e_d}{6} (\mu^2 (3\eta^2 + 3\eta(\bar{\gamma} - \underline{\gamma}) + (\bar{\gamma} - \underline{\gamma})^2) - (\bar{\gamma} - \underline{\gamma})^2) = M$$

and all types participate.

For $M > \frac{1}{2} e_d \eta (\eta + \bar{\gamma} - \underline{\gamma})$ we have $\mu(M) = 1$.

Proof. See appendix A.3. $\square$

Lemma 6. Let $\mu(M) := \frac{1}{\lambda(M)}$, where $\lambda(M)$ denotes the global institution's shadow value of funds when its budget is M . The optimal mechanism can be implemented by the following reward function:

$$t(\tau) = \alpha \max(0, \tau - \tau^*) + \beta (\max(0, \tau - \tau^*))^2 \quad (1)$$

where α , β and τ^* are dermined from the budget M via the following:

For $M \leq \frac{e_d (\bar{\gamma} - \underline{\gamma})^2 (2 + \frac{\bar{\gamma} - \underline{\gamma}}{\eta})}{6(1 + \frac{\bar{\gamma} - \underline{\gamma}}{\eta})^2}$ we have:

$$\alpha = 0, \beta = \frac{1 - \mu(M)}{2(2 - \mu(M))} e_d, \tau^* = \bar{\gamma} - \frac{\eta \mu(F)}{1 - \mu(F)}$$

For $M \geq \frac{e_d \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}$ we have:

$$\alpha = \sqrt{\frac{(\eta \mu(M) - (\bar{\gamma} - \underline{\gamma})(1 - \mu(M)))^2}{2 - \mu(F)}} e_d, \beta = \frac{1 - \mu(M)}{2(2 - \mu(M))} e_d, \tau^* = \frac{\eta \mu - \sqrt{(2 - \mu)((\bar{\gamma} - \underline{\gamma})(1 - \mu) - \mu \eta)^2 + (1 - \mu) \bar{\gamma}}}{1 - \mu}$$

which also writes as follows:

$$\alpha = \frac{(\eta \mu(M) - (\bar{\gamma} - \underline{\gamma})(1 - \mu(M)))}{\sqrt{2 - \mu(M)}} e_d, \beta = \frac{1 - \mu(F)}{2(2 - \mu(F))} e_d, \tau^* = \bar{\gamma} - \sqrt{(2 - \mu)(\bar{\gamma} - \underline{\gamma})} +$$

$$\frac{\mu(1 + \sqrt{(2 - \mu)})}{1 - \mu} \eta$$

In particular, we have:

β is strictly decreasing in M .

α is strictly increasing in M .

τ^* is first strictly decreasing in M for $M \in [0, e_d (\bar{\gamma} - \underline{\gamma})^2 (2 + \frac{\bar{\gamma} - \underline{\gamma}}{\eta})]$ and increasing in M for $M > [e_d (\bar{\gamma} - \underline{\gamma})^2 (2 + \frac{\bar{\gamma} - \underline{\gamma}}{\eta}), \frac{1}{2} e_d \eta (\eta + \bar{\gamma} - \underline{\gamma})]$.

Proof. See appendix A.4 $\square$

2.3 The limiting case where the global institution only cares about reducing emissions

In the general model studied so far, the countries' utilities (excluding climate change effects) enter the global institution's Lagrangian in two ways: Firstly, it appears in its objective function, since the global institution cares intrinsically about it. Secondly, it enters through the participation and incentive compatibility constraints. How do the results change if the first effect disappears, i.e. if the global institution's objective function only consists of the emissions reductions?

It turns out that the optimal mechanism changes very little, at least if the available budget is not much larger than what is plausible. I show this in section 3.4. Given this fact, it is interesting to study the limiting case where the global institution only cares about emissions reductions. The following Lemma gives explicit expressions for the optimal mechanism under this assumption:

Lemma 7. *Consider the limiting case where the global institution only cares about reducing emissions, which corresponds to the limit as the social cost of carbon goes to ∞ . Then we have at the optimal mechanism:*

For $M \leq \frac{e_d(\bar{\gamma}-\underline{\gamma})^2}{3}$ the set of participating types is given by $[\bar{\gamma} - (\frac{3(\bar{\gamma}-\underline{\gamma})M}{e_d})^{\frac{1}{3}}, \bar{\gamma}]$ and these types choose the following tax rates:

$$\tau(\gamma) = \gamma + \left(\frac{3(\bar{\gamma} - \underline{\gamma})M}{e_d} \right)^{\frac{1}{3}} - (\bar{\gamma} - \gamma)$$

The resulting emissions reductions are:

$$\frac{e_d}{2} \left(\frac{3M}{e_d} \right)^{\frac{2}{3}} (\bar{\gamma} - \underline{\gamma})^{-\frac{1}{3}}$$

Thus the ratio of the resulting emissions reductions to the emissions reductions achievable under complete information is:

$$\frac{3^{\frac{2}{3}}}{2^{\frac{3}{2}}} \left(\frac{M}{e_d} \right)^{\frac{1}{6}} (\bar{\gamma} - \underline{\gamma})^{-\frac{1}{3}}$$

For $M \geq \frac{e_d(\bar{\gamma}-\underline{\gamma})^2}{3}$ all types participate and choose the following tax rates:

$$\tau(\gamma) = \gamma + \sqrt{2\frac{M}{e_d} + \frac{1}{3}(\bar{\gamma} - \underline{\gamma})^2} - (\bar{\gamma} - \gamma)$$

The resulting emissions reductions are:

$$e_d \left(\sqrt{2\frac{M}{e_d} + \frac{1}{3}(\bar{\gamma} - \underline{\gamma})^2} - \frac{(\bar{\gamma} - \underline{\gamma})^2}{2} \right)$$

Thus the ratio of the resulting emissions reductions to the emissions reductions achievable under complete information is:

$$\frac{\sqrt{2e_dM + \frac{1}{3}e_d(\bar{\gamma} - \underline{\gamma})^2 - e_d\frac{(\bar{\gamma} - \underline{\gamma})^2}{2}}}{\sqrt{2e_dM}} = \sqrt{1 + \frac{1}{3M}(\bar{\gamma} - \underline{\gamma})^2 - \sqrt{e_d}\frac{(\bar{\gamma} - \underline{\gamma})^2}{2\sqrt{2M}}}$$

In particular, this ratio converges³ to 1 as M goes to ∞ .

Lemma 8. Consider the limiting case where the global institution only cares about reducing emissions, which corresponds to the limit as the social cost of carbon goes to ∞ . The optimal mechanism can be implemented by the following reward function:

$$t(\tau) = \alpha \max(0, \tau - \tau^*) + \beta (\max(0, \tau - \tau^*))^2 \quad (2)$$

where α , β and τ^* are determined from the budget M via the following:

For $M \leq \frac{e_d(\bar{\gamma} - \underline{\gamma})^2}{3}$ we have:

$$\alpha = 0, \beta = \frac{1}{4}e_d, \tau^* = \bar{\gamma} - \left(\frac{3(\bar{\gamma} - \underline{\gamma})}{e_d}M\right)^{\frac{1}{3}}$$

For $M \geq \frac{e_d(\bar{\gamma} - \underline{\gamma})^2}{3}$ we have:

$$\alpha = \frac{(\frac{3(\bar{\gamma} - \underline{\gamma})}{e_d}M)^{\frac{1}{3}} - (\bar{\gamma} - \underline{\gamma})}{\sqrt{2}}e_d, \beta = \frac{1}{4}e_d, \tau^* = \bar{\gamma} - \sqrt{2}(\bar{\gamma} - \underline{\gamma}) + \left(\frac{3(\bar{\gamma} - \underline{\gamma})}{e_d}M\right)^{\frac{1}{3}}(1 + \sqrt{2})$$

Proof. See appendix A.5. $\square$

2.4 The optimal mechanism with 2 fossil fuels

So far we have normalized the units of the fossil fuel such that total fossil fuel consumption in the hypothetical scenario where all countries choose $\tau = 0$ is 1. At the actual status quo, where the global institution studied here does not yet exist, each country chooses $\tau = \gamma$. The following lemma gives us the resulting fossil fuel use:

Lemma 9. For $M = 0$ we have for the aggregate use of the fossil fuel: $\int_{\gamma=\gamma_L}^{\gamma_H} f(\gamma)y(\tau(\gamma))d\gamma = 1 - \frac{e_d}{2}(\bar{\gamma} + \underline{\gamma})$.

The preceding lemma also tells us the aggregate spending on fossil fuels, since we have normalised the price to $p = 1$.

Now let us consider the case where there are two fossil fuels, coal and oil. Suppose that the model studied so far can be applied to each of these separately. Moreover, let us now introduce two parameters, s_{coal} and s_{oil} that simply scale up the entire model. Specifically, define s_{coal} such that in the absence of the global institution total spending on coal is s_{coal} and similarly for s_{oil} .

The following Lemma informs the case where the available budget does not optimally lead to full participation, which is empirically relevant, as we will see.

³The actual ratio turns out to be less than 0.6 even for the optimistic assumption that $M = \$100$ billion per year, as I show in section 3. In fact, the rate of convergence is ridiculously slow.

Lemma 10. Suppose the global institution can split its exogenous budget M between two funds, one for optimally contracting on carbon pricing for coal and one for optimally contracting on carbon pricing for oil. Suppose that $M \leq \frac{s_{coal}}{1 - \frac{e_{d coal}}{2}(\bar{\gamma}_{coal} + \underline{\gamma}_{coal})} \frac{e_{d coal} \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2} + s_{oil} \frac{e_{d oil} \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}$. Let $(M_{coal}(M), M_{oil}(M))$ be the optimal way to split the budget between the two funds, assuming that both global coal supply and global oil supply are perfectly elastic. If both oil supply and coal supply are perfectly elastic then we have:

$$\frac{M_{coal}(M)}{M_{oil}(M)} = \frac{s_{coal}}{s_{oil}} \frac{1 - \frac{e_{d oil}}{2}(\bar{\gamma}_{oil} + \underline{\gamma}_{oil})}{1 - \frac{e_{d coal}}{2}(\bar{\gamma}_{coal} + \underline{\gamma}_{coal})} \frac{\bar{\gamma}_{oil} - \underline{\gamma}_{oil}}{\bar{\gamma}_{coal} - \underline{\gamma}_{coal}} \frac{e_{d coal}}{e_{d oil}} \left(\frac{\eta_{coal}}{\eta_{oil}} \right)^3$$

3 Numerical applications

For all of this section I will assume that the only countries eligible for receiving the reward payments from the global institution are the non-annex 1 countries. I show results for annual budgets ranging from \$0 to \$100 billion. The rich countries had promised to provide annual funding for \$100 billion to the non-annex 1 countries from 2015 onwards for mitigation and adaptaion. However, current annual funding for all global environmental institutions is only 6 billion (ref).

3.1 Empirical calibration of parameters

$e_{d coal} = 0.7$, based on Keen et al. (2019) for the price elasticity of coal demand

$e_{s coal} = 1.3$, based on Dahl (2009) for the price elasticity of coal supply

$\eta_{coal} = 14.7$ based on a social cost of carbon of \$417 per ton of CO2 (based on Ricke et al. (2018)).⁴

$s_{coal} = 575$ billion for the annual dollar amount of spending on coal by non-annex 1 countries

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⁴This paper tries to answer a normative question: How should the global institution best be designed so as to maximise aggregate well-being? Therefore, it is appropriate to use a 0 rate of pure time preference when aggregating future climate damages.

Recall that by Lemma 1 our demand specification has the following implicit normalisation: Price at the status quo is 1. Thus η is the ratio of the social cost of a unit of coal divided by its price. To compute this we use the following:

25 million Btu per short ton by: https://openei.org/wiki/Definition:Bituminous_coal#:~:text=Bulk%20density%20typically%20runs%20to,mineral%2Dmaterial%2Dfree%20basis:

79kgCO2 per GJ by https://www.engineeringtoolbox.com/co2-emission-fuels-d_1085.html
79/0.94791 kgCO2 per million Btu by <http://convert-to.com/conversion/energy/convert-gj-to-btu.html>

Hence 79/0.94791*25kgCO2 per short ton
price p=\$58.93 per short ton by <https://www.eia.gov/energyexplained/coal/prices-and-outlook.php>

A social cost of carbon of 417\$ per ton of CO2 leads to $\eta/p = 417 * 231.7/1000 * 25 = 14.7$

⁵719.2\$bn is Global expected revenue from Coal Mining in 2021 by <https://www.ibis-world.com/global/market-size/global-coal-mining/>. Assuming that 80% of this is burnt in non-annex 1 countries (to be replaced by a proper estimate),

$e_{oil} = 0.5$, based on Keen et al. (2019) for the price elasticity of oil demand
 $e_s = 0.32$, based on Golombek et al. (2018) for the price elasticity of oil supply

$\eta_{oil} = 2.78$ based on a social cost of carbon of \$36 per ton of CO2 (based on Ricke et al. (2018)).⁶

$s_{oil} = 1148$ billion for the annual dollar amount of spending on oil by non-annex 1 countries

$\gamma_{coal} = 0, \bar{\gamma}_{coal} = 0.22$, based on the observation that coal subsidies are very close to 0 globally (Coady et al. 2017, Coady et al. 2019) and based on the observation that some developping countries have modest carbon prices covering coal, most notably China with \$6.3 per ton of CO2 in 2020 , which corresponds in the model to China having $\gamma = 0.22$.

$\gamma_{oil} = -1, \bar{\gamma}_{oil} = 0.1$, based on the following data from Sovacool (2017):

⁶Recall that by Lemma 1 our demand specification has the following implicit normalisation: Price at the status quo is 1. Thus η is the ratio of the social cost of a unit of coal divided by its price. To compute this we use the following: The EPA states: The average carbon dioxide coefficient of distillate fuel oil is 429.61 kg CO2 per 42-gallon barrel (EPA 2018). The fraction oxidized to CO2 is 100 percent (IPCC 2006). The eia states: The price of Brent crude oil, the international benchmark, averaged \$64 per barrel (b) in 2019. Assuming a social cost of carbon of \$417 per ton of CO2, we get: $\eta/p = 417 * 0.42961/64 = 2.78$.

⁷world oil consumption= 98.27M bbl/d for 2019 by https://ycharts.com/indicators/world_oil_consumption

The eia states: The price of Brent crude oil, the international benchmark, averaged \$64 per barrel (b) in 2019.

This gives that global oil consumption is \$2.296 trillion per year (in 2019)

The share of that being consumed in non-annex 1 countries is 50% (a guess, to be updated with data).

Figure 1:

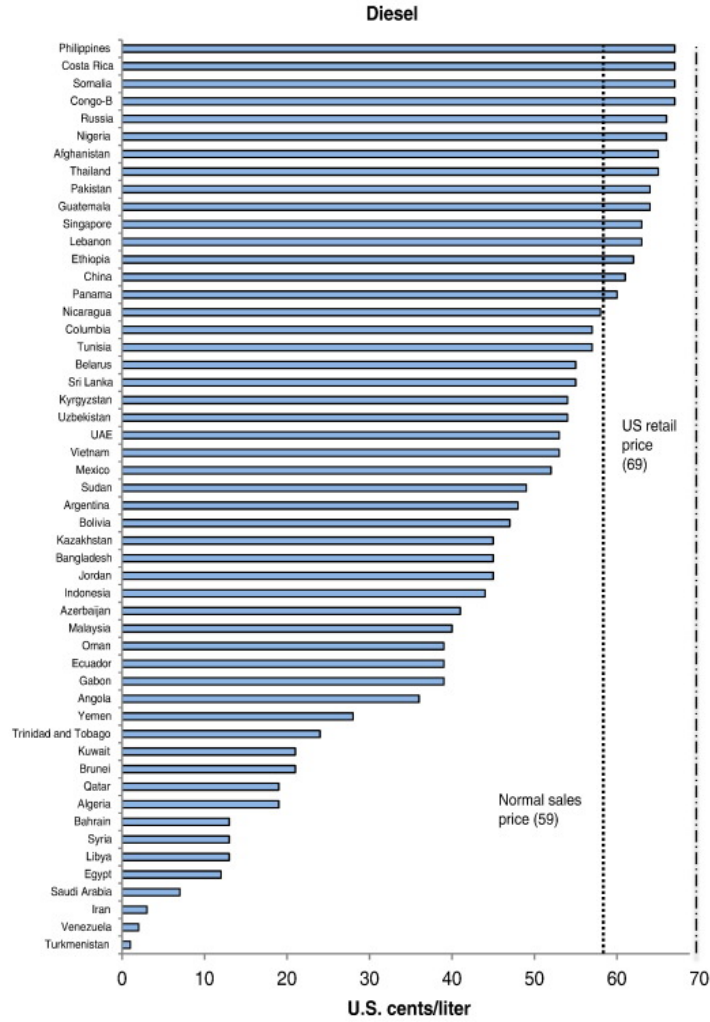
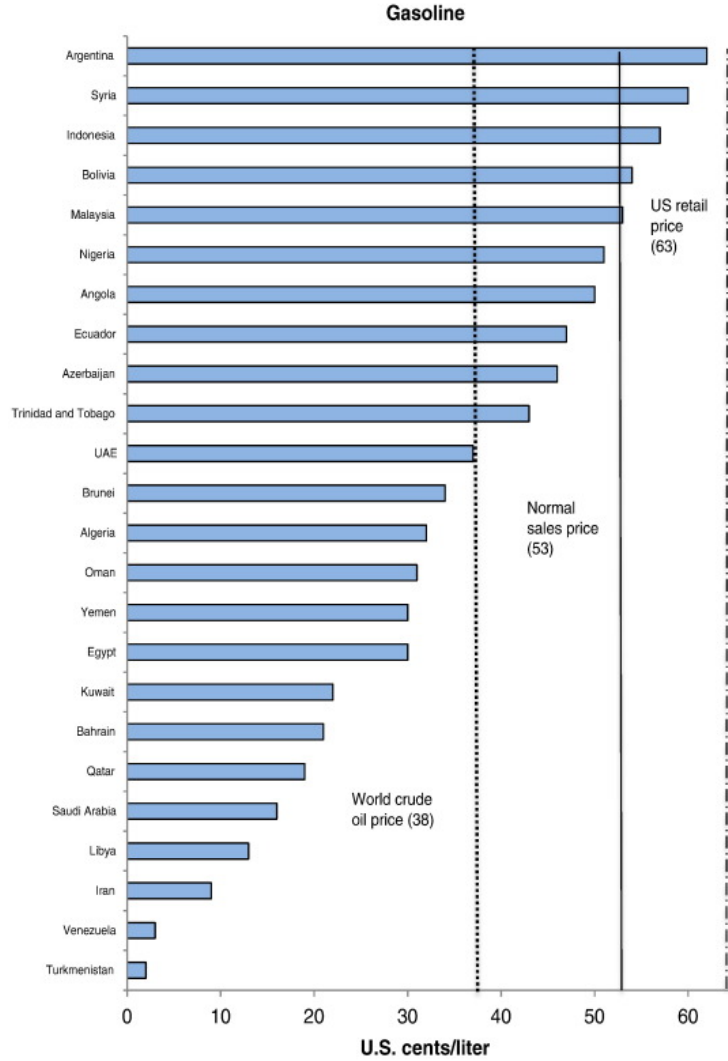


Figure 2:



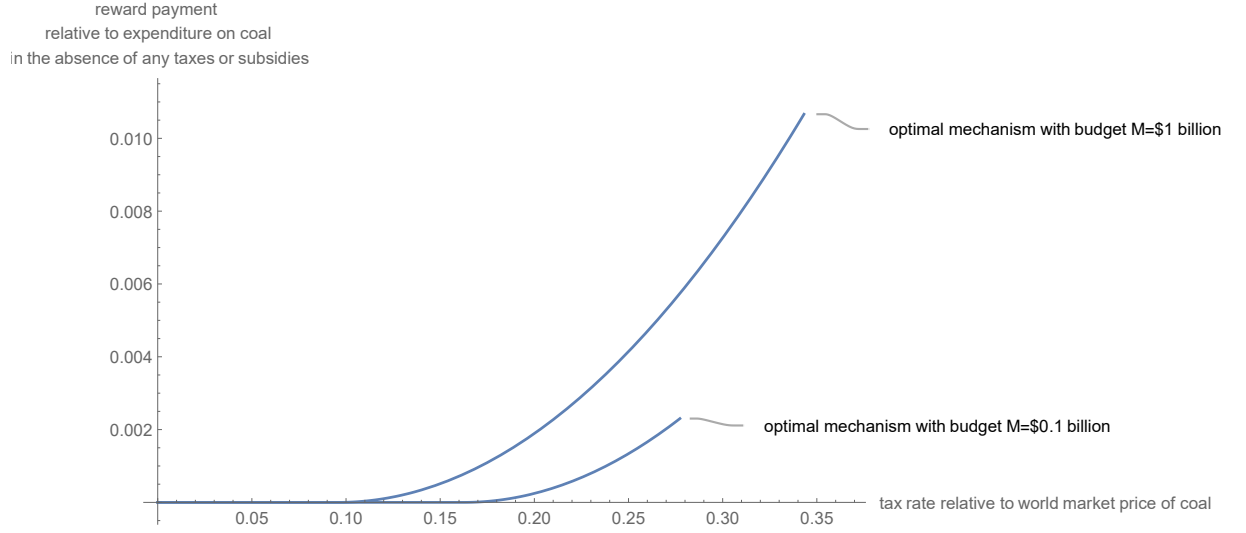
3.2 The optimal design of a global Fund for rewarding carbon pricing of coal in non-Annex 1 countries

By Lemma 5, the minimal annual budget leading to full participation at the optimal mechanisms is:

$$\frac{s_{coal}}{1 - \frac{e_{d\ coal}}{2}(\bar{\gamma}_{coal} + \gamma_{coal})} \frac{e_{d\ coal} \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2} = \$5.6billion$$

For annual budgets smaller than this, the optimal reward payment function is tangent to the x-axis at the minimal participating type, by Lemma , and as seen in the following plot:

Figure 3: optimal reward payment functions for carbon taxes on coal with annual budget of \$0.1 billion and \$1 billion



For budgets larger than \$5.6billion, there is full participation at the optimal mechanism.

Figure 4:

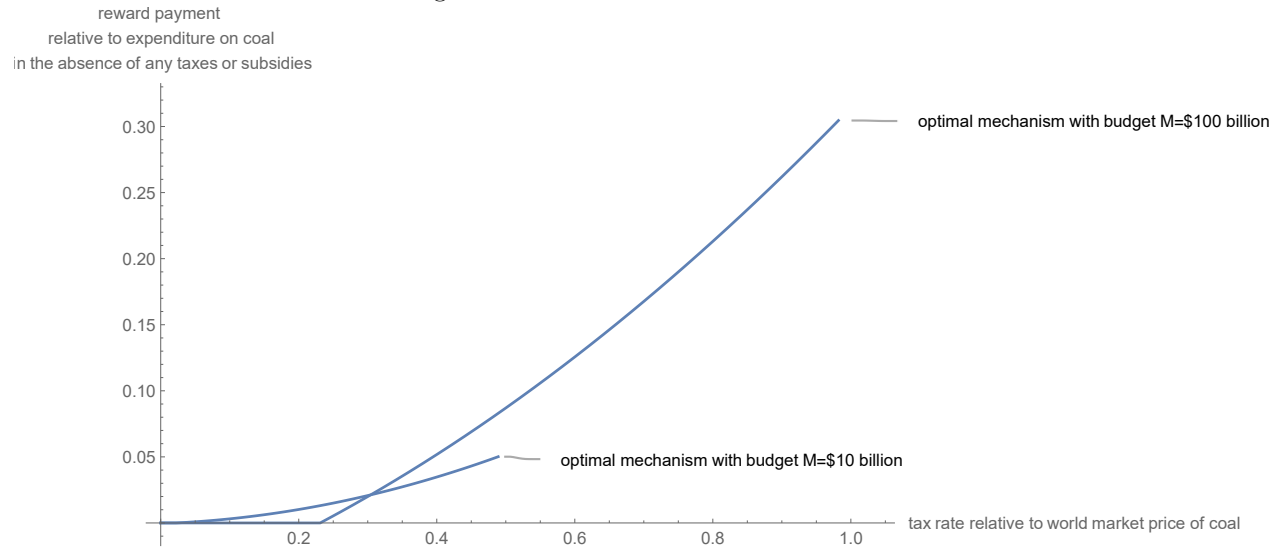


Figure 5:

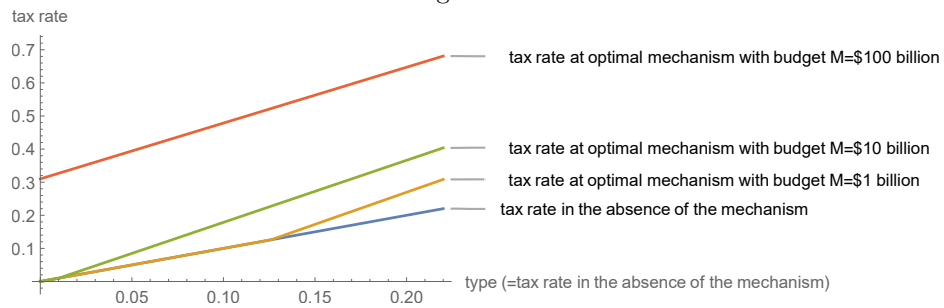


Figure 6: coal emissions reductions under complete and incomplete information

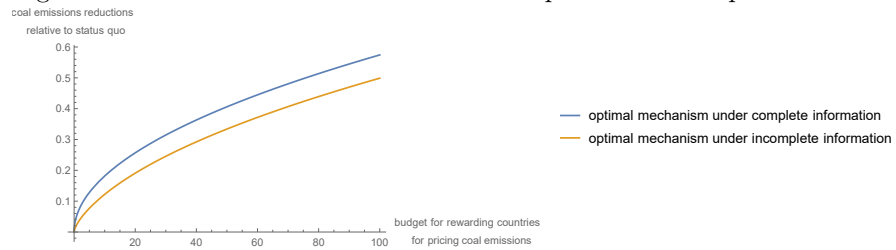
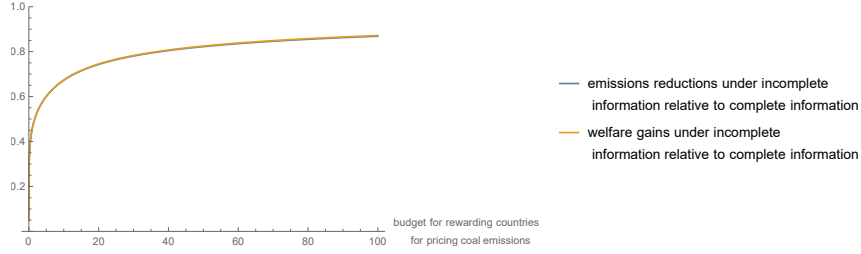
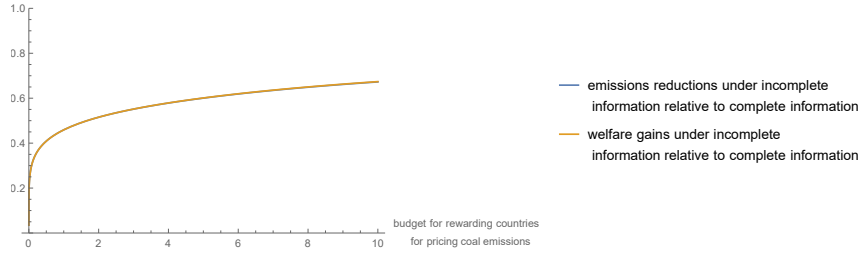


Figure 7: welfare gains and emissions reduction under incomplete information relative to complete information



Zooming in:

Figure 8: welfare gains and emissions reduction under incomplete information relative to complete information



3.3 The optimal design of a global Fund for rewarding carbon pricing of oil in non-Annex 1 countries

By Lemma 5, the minimal annual budget leading to full participation at the optimal mechanisms is:

$$\frac{\frac{s_{coal}}{1 - \frac{e_{d\ coal}}{2}(\bar{\gamma}_{coal} + \underline{\gamma}_{coal})}}{e_{d\ coal}\eta(\bar{\gamma} - \underline{\gamma})^2(2\eta + \bar{\gamma} - \underline{\gamma})} = \$58.2billion$$

Figure 9:

reward payment relative to expenditure on oil in the absence of any taxes or subsidies

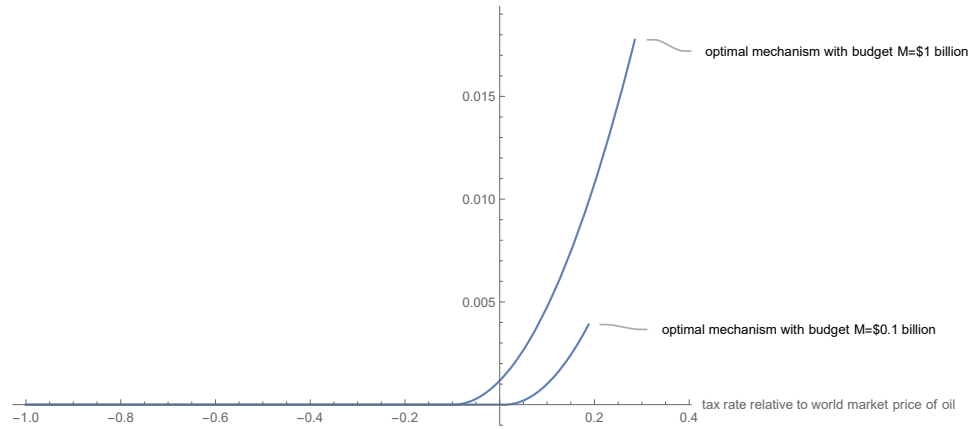
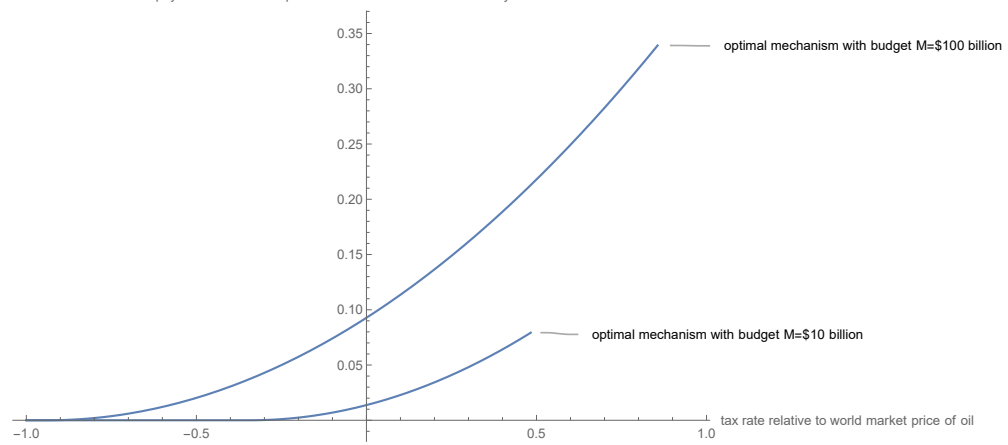


Figure 10:

reward payment relative to expenditure on oil in the absence of any taxes or subsidies



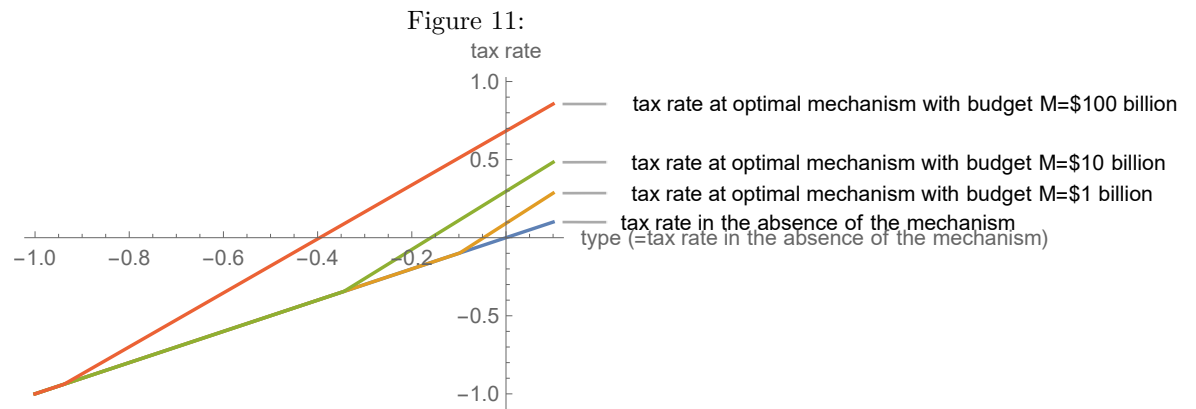


Figure 12: oil emissions reductions under complete and incomplete information

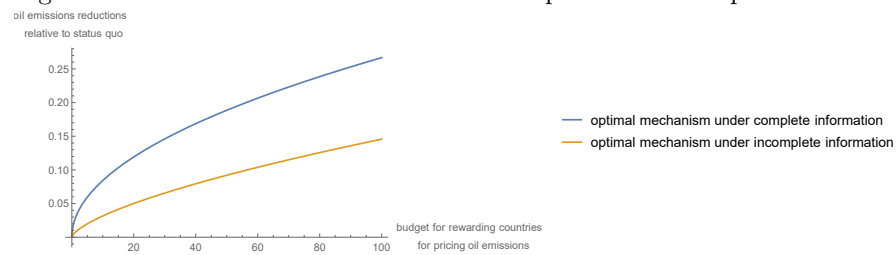
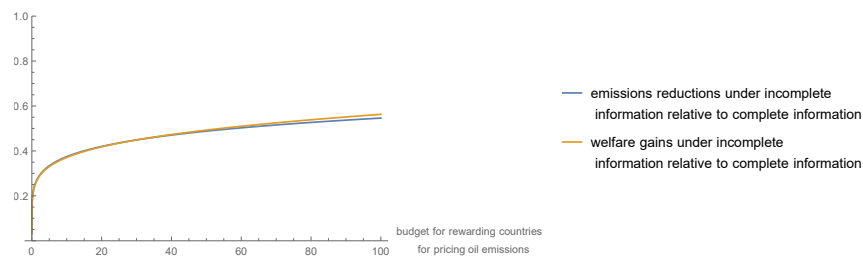
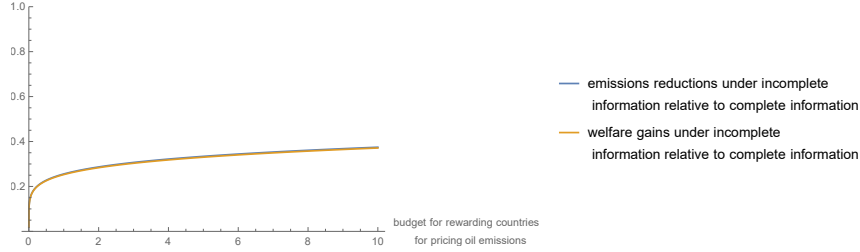


Figure 13: welfare gains and emissions reduction under incomplete information relative to complete information



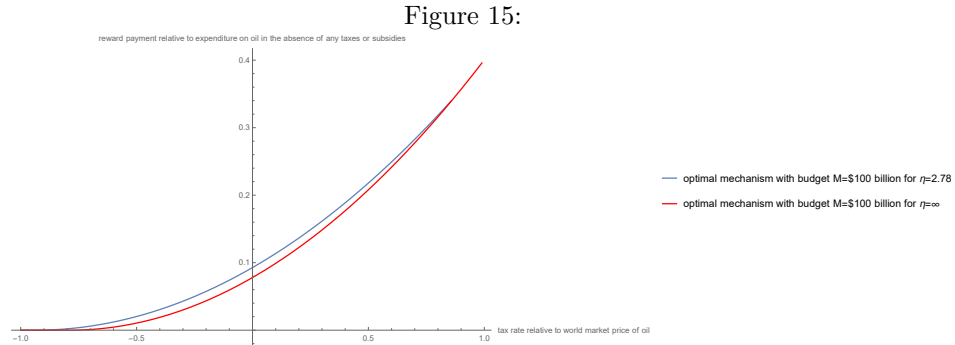
Zooming in:

Figure 14: welfare gains and emissions reduction under incomplete information relative to complete information



3.4 Sensitivity analysis with respect to the social cost of carbon η

It turns out that if the global institution only cares about reducing global emissions (corresponding to $\eta = \infty$), the results presented in the previous sections hardly change at all unless the available budget is extremely large. Specifically, for budgets up to \$10 billion the optimal reward payment functions are almost undistinguishable. For coal this is true even for a budget of \$100 billion. For oil, a slight difference emerges: the following graph shows that for an annual budget of \$100 billion, the global institution concentrates its incentive power a bit more on the high types if it only cares about reducing global emissions:



This can be explained as follows: If the global institution cares intrinsically about the current utility (excluding climate change damages) of the countries that it contracts with then it is somewhat concerned about spreading out effort (i.e. increases in the carbon taxes), so as to achieve a given aggregate emissions reductions at lower costs.

3.5 The optimal split of a given overall budget between a between the coal carbon pricing reward fund and the oil carbon pricing reward fund

For simplicity, let us assume

Applying Lemma 10 yields for the optimal budget split:

$$\frac{M_{coal}(M)}{M_{oil}(M)} = \frac{s_{coal}}{s_{oil}} \frac{1 - \frac{e_{d,oil}}{2}(\bar{\gamma}_{oil} + \gamma_{oil})}{1 - \frac{e_{d,coal}}{2}(\bar{\gamma}_{coal} + \gamma_{coal})} \frac{\bar{\gamma}_{oil} - \gamma_{oil}}{\bar{\gamma}_{coal} - \gamma_{coal}} \frac{e_{d,coal}}{e_{d,oil}} \left(\frac{\eta_{coal}}{\eta_{oil}} \right)^3$$

4 Implementing the mechanism in practice

I now propose an operational method for implementing carbon pricing reward funds in practice, using this paper’s model to define a reward payment function. The method is based on four design choices that I detail in the following sub-sections.

4.1 Using the limiting case where the global institution only cares about reducing emissions

In section 3.4, I reported on results showing that if the global institution only cares about reducing emissions then the optimal mechanism will hardly different from the case where the global institution weighs emissions reduction against current day welfare using the social cost of carbon estimates explained in section 3.1. The former assumption has the advantage of making the optimal mechanism much easier understandable, as shown by Lemma 7 and Lemma 8 that give explicit formulae for the coefficients of the optimal reward payment function. I therefore propose to use this assumption.

4.2 Creating a separate carbon pricing reward fund for each type of fossil fuel

In the numerical applications of section 3 I considered the case where there is a separate carbon pricing reward fund for coal and oil. In practice, I propose to further differentiate and to instead create a separate carbon pricing reward fund for each sub-type of fuel (oil used for heating, oil used for transportation, coal used for electricity generation, coal used for steel production etc.). This architecture is likely to both lead to easier implementation and greater welfare gains, as I will now briefly argue.

In practice, countries differentiate their fossil fuel taxes and subsidies according to the sub-type of fuel. Under the architecture with separated carbon pricing reward funds proposed here, the model can be directly applied. If instead a single carbon pricing reward fund was used for all types of coal or oil, then application of the model would require some way of aggregating the different subsidies/taxes across the different sub-types.

Concerning efficiency, Lemma 8 shows that the optimal reward payment function depends strongly on the elasticity of demand for the sub-type of fuel.

These elasticities vary substantially across sub-types of fuel (ref). This suggests that lumping several sub-types of fuel together in a single carbon pricing reward fund could lead to significant welfare losses.

4.3 Allowing donors to earmark monetary contributions to each of the carbon pricing reward funds

For a given amount of funding, Lemma 10 provides the optimal amount to split it between the different carbon pricing reward funds. This suggests a centralised architecture where donations could only be made to a centralised institution which would then split it across the different carbon pricing reward according to the formula. However, I will now argue in favor of the separated architecture that I propose here, where donors can earmark their contributions to the separate funds.

An advantage of the separated architecture is that it enhances donors incentives to contribute funding. Specifically, oil importers benefit from lower world oil prices. They therefore benefit particularly from donations to the oil carbon pricing reward funds, since these lower the world market price of oil. In Stern (2021) I find in a related model that this incentive effect always dominates the efficiency losses that come from sub-optimal splitting of funding across different funds.

4.4 Estimating the emissions in the hypothetical situation without any taxes or subsidies

In the model, the transfer to a country is the product of the reward function (which depends on the carbon price only) and the expenditure on the fossil fuel in the hypothetical scenario where the country sets a 0 carbon price. In the model this latter factor is 1 by our choices of normalization. In practice, it would be necessary to instead estimate the expenditure on the fossil fuel in the hypothetical scenario where the country sets a 0 carbon price times.

It seems feasible to define a standardized methodology for this. It should take into account not only the explicit carbon prices but also other policies that reduce emissions. The goal for the methodology should be to generate as accurate as estimate as possible of the expenditure on the fossil fuel that would arise in the absence of any government intervention. This would be important to ensure that the mechanism does not in any way undermine countries' incentives to pursue other policies to reduce emissions.

4.5 Ensuring budget balance

In the model, the global institution can perfectly anticipate the changes in countries' tax rates that it induces through the reward payments. Thus the global institution can make sure that it always ends up spending exactly its budget. In practice, this will not hold, so the mechanism needs to be modified.

I propose the following approach: Before the beginning of each year, the optimal mechanism is computed in the model presented here, using best available estimates of price elasticities of demand for the fossil fuel. If the mechanism thus obtained involves partial participation, then proceed as follows: announce the reference level τ^* and commit to it. Recall that by Lemma 8 the optimal reward payment function in this case is simply:

$$t(\tau) = \beta(\max(0, \tau - \tau^*))^2 \quad (3)$$

Now the value of β would be determined for the year in question by the budget balance condition. As a result, when a country chooses its carbon tax at the beginning of the year, it would certainty about whether it will get any reward payment or not. How much reward payments it will receive will depend on how large β ends up being, which depends on the other countries' choices.

Now consider the case where the mechanism computed to be optimal according to the model involves full participation. Recall that by Lemma 8 the optimal reward payment function in this case is:

$$t(\tau) = \alpha \max(0, \tau - \tau^*) + \beta(\max(0, \tau - \tau^*))^2 \quad (4)$$

with positive α and β . Again, before the beginning of the year, the global institution announces the value of τ^* it has computed. It commits to it. The ratio $\frac{\alpha}{\beta}$ would be fixed according to values outputted by the model. (By Lemma 7, we have $\frac{\alpha}{\beta} = 2\sqrt{2}((\frac{3(\bar{\gamma}-\underline{\gamma})}{e_d}M)^{\frac{1}{3}} - (\bar{\gamma} - \underline{\gamma}))$.) The absolute values of α and β would then be set at the end of the year (when the reward payments for that year are actually made) so as to exactly achieve budget balance.

The approach I have defined here has the advantage of avoiding cases of disappointment where a country hopes to get reward payments but ends up not getting any. This is ensured by having the global institution commit to the value of τ^* that it uses for a given year before that year begins. Of course this means that all the uncertainty about the actual reward payments resides in the rates α and β at which countries are rewarded. However, it seems better to concentrate the uncertainty at the intensive rather than the extensive margin.

5 Design questions

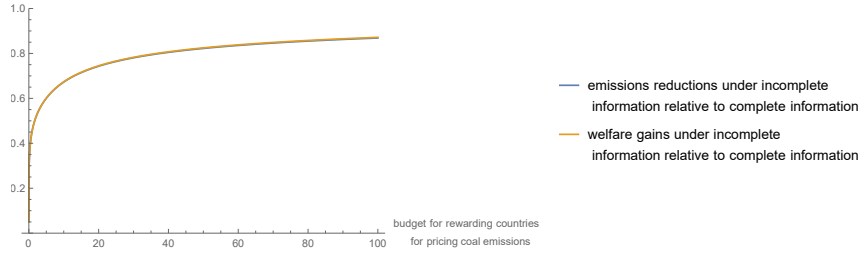
Having proposed in section 4 a concrete procedure for defining separate carbon pricing reward funds for each sub-type of fossil fuel, I now discuss alternative possible design choices in light of the results from section 3.

5.1 Should the global institution condition reward payments on other variables than the current tax rates?

Alternatively to the simple kind of contracting studied in this paper, the global institution could condition its reward payments on other variables, such as past carbon prices that the country had in place. The advantage of this would be that

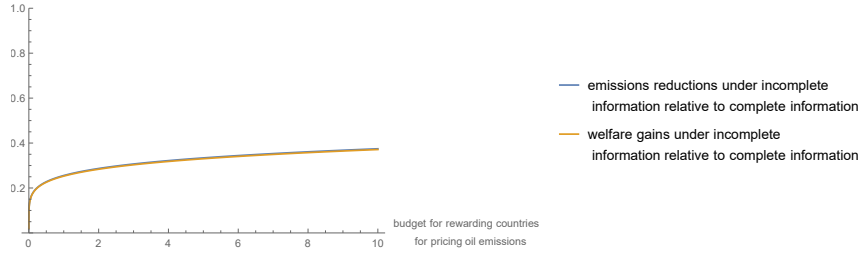
the global institution could reduce the informational rents that countries reap. The model with complete information (section 2.1) provides an upper bound on the gains that could be realised from this. For coal, the results from section 3.2 suggest that the gains from eliminating informational rents are rather limited, at least for when the budget is large:

Figure 16: welfare gains and emissions reduction under incomplete information relative to complete information



However, in the case of oil, the gains are substantial if the budget is low:

Figure 17:



However, even in this case it is unclear whether it would be good overall to condition the reward payments on other variables than the current carbon pricing. This is because it would inadvertently retroactively reward countries for globally detrimental past policies. To see why, consider the case where reward payments are conditioned on carbon prices at a time in the past before the carbon pricing funds' creation. This would allow for a reduction in rents: Current preferred carbon prices are likely to be strongly correlated to past carbon prices. Thus the reference levels τ^* could be made an increasing function of past carbon prices, thereby cutting rents for the high types.

As long as the reference levels are conditioned on actions (e.g. carbon prices) taken before the introduction of the new carbon pricing reward funds, there would be no direct distortion of incentives. However, this would set a bad precedent. Reward payment schemes of the kind discussed here might be able to play a valuable role in other areas of international collective action in the

future. For example, it might be important to create analogous schemes for rewarding countries for their spending on disease surveillance through metagenomic sequencing so that the world will stand a better chance to extinguish future potential pandemics before they spread (ref).

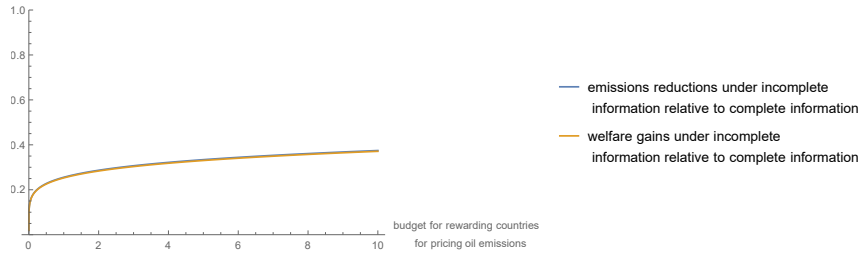
If the carbon pricing reward payment funds were to effectively reward countries for past low carbon prices (e.g. by making the reference levels τ^* an increasing function of past carbon prices, as discussed in the preceding paragraph), then this would create the (valid) expectation that this will again occur in other realms such as pandemic prevention. As a result, it would create perverse incentives.

5.2 At what level of funding would the creation of a carbon pricing reward fund be valuable?

So far, global environmental institutions such as the Clean Development Mechanism (CDM) and the Green Climate Fund have relied on project based approaches to climate change mitigation. An advantage of project based approaches is that they can reduce informational rents. For example, the CDM only certifies projects that it assesses would not have been implemented in the absence of the financing that it provides. It pays projects proportionally to the amount of emissions reductions that they achieve. If one assumes that the marginal abatement cost curve is linear, then one finds that the CDM ends up paying half its budget as informational rents and using its other half as if it was contracting optimally under complete information.

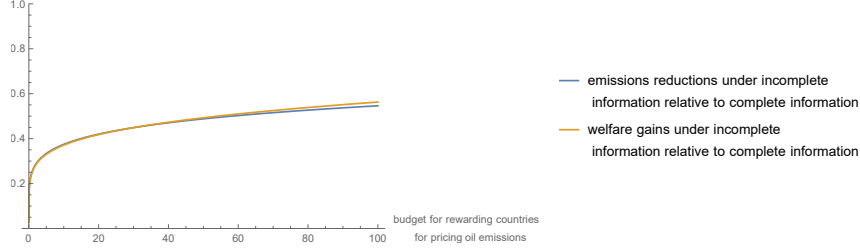
For low annual budgets, the carbon pricing reward funds achieve much lower emissions reductions, particularly in the case of oil, as the results from section 3 shows:

Figure 18:



However, for larger annual funding levels, the carbon pricing reward funds would achieve larger welfare gains, even in the case of oil:

Figure 19:



If the world were to mobilise additional funding for global climate change institutions in the order of the existing annual amounts (5 billion), then it might be optimal to direct this additional funding to newly established carbon pricing reward funds.

There are further considerations supporting this conclusion: For the carbon pricing reward funds, the main administrative costs are roughly fixed: The countries' carbon prices need to be measured and the counterfactual fossil fuel use in the counterfactual scenario without any carbon taxes need to be estimated (see section 4.4). For project based approaches like the Clean Development Mechanism and the Green Climate Fund, the administrative costs are roughly proportional to the number of projects.

6 Conclusion and limitations

This paper proposes a simple contract theory model. It provides simple explicit formulae for the optimal way for a global institution to reward countries for taxing their fossil fuels. Based on it, I have proposed a procedure for determining a reward payment function in practice.

A major limitation of the procedure is that it does not allow for learning over time. If and when the mechanism is implemented, we can observe how countries actually respond to the reward payment functions. It is unlikely that this will correspond very closely to the prediction of the model. It seems important to develop more flexible contract theory models for constructing procedures for adjusting the reward payment function over time in light of the observed changes in tax rates.

A Proofs for section

A.1 Proof of Lemma 3

Proof. Let L denote the global institution's Lagrangian and λ the Lagrange multiplier associated with the budget constraint:

$$L = \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(v(\tau(\gamma), \gamma) - \eta y(\tau(\gamma)) - \lambda t(\tau(\gamma))) d\gamma$$

From now on let us focus on the case where M is sufficiently small for the voluntary participation constraints to all be binding. We get:

$t(\tau(\gamma)) = -v(\tau(\gamma), \gamma)$ and thus:

$$L = \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(v(\tau(\gamma), \gamma) - \eta y(\tau(\gamma)) + \lambda v(\tau(\gamma))) d\gamma$$

The first order condition is:

$$\frac{\partial v}{\partial \tau}(\tau(\gamma), \gamma) - \eta y'(\tau) + \lambda \frac{\partial v}{\partial \tau}(\tau(\gamma), \gamma) = 0$$

Now using that $y(\tau) = -pe_d - e_d\tau + e_d + 1$ and $v(\tau(\gamma), \gamma) = -\frac{1}{2}e_d(\tau - \gamma)^2$, we obtain:

$$-(\lambda + 1)e_d(\tau - \gamma) + \eta e_d = 0$$

Thus the optimal contract is given by:

$$\tau(\gamma) = \gamma + \frac{\eta}{\lambda + 1}$$

Integrating yields:

$$M = \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} t(\gamma) d\gamma = \frac{e_d}{2} \left(\frac{\eta}{\lambda + 1} \right)^2$$

By eliminating the Lagrange multiplier λ , we obtain the claimed results. $\square$

A.2 Proof of Lemma 4

Proof. Let L denote the global institution's Lagrangian and λ the Lagrange multiplier associated with the budget constraint:

$$\begin{aligned} L &= \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(v(\tau(\gamma), \gamma) - y(\tau)\eta - \lambda(-u(\tau(\gamma), \gamma) + U(\gamma))) d\gamma \\ L &= \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(-y(\tau)\eta - \lambda U(\gamma) + (\lambda + 1)v(\tau(\gamma), \gamma)) d\gamma \end{aligned} \quad (5)$$

Using integration by parts, we get:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)U(\gamma) d\gamma = [-(1 - F(\gamma))U(\gamma)]_{\gamma=\underline{\gamma}}^{\bar{\gamma}} + \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} (1 - F(\gamma))U'(\gamma) d\gamma$$

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)U(\gamma)d\gamma = U(\underline{\gamma}) + \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} (1 - F(\gamma))U'(\gamma)d\gamma$$

Now using that $U'(\gamma) = y(\gamma) - y(\tau(\gamma))$, we get:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)U(\gamma)d\gamma = U(\underline{\gamma}) + \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} (1 - F(\gamma))(y(\gamma) - y(\tau(\gamma)))d\gamma$$

Substituting this into equation 5 yields:

$$L = \lambda U(\underline{\gamma}) + \int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} f(\gamma)(-y(\tau)\eta - \lambda \frac{1 - F(\gamma)}{f(\gamma)}(y(\gamma) - y(\tau(\gamma))) + (1 + \lambda)v(\tau(\gamma), \gamma))$$

The first order condition for τ is thus:

$$-y'(\tau)\eta + \lambda \frac{1 - F(\gamma)}{f(\gamma)}y'(\tau(\gamma)) + (1 + \lambda)\frac{\partial v}{\partial \tau}(\tau, \gamma) = 0$$

Now using that $y'(\tau) = -e_d$ and $\frac{\partial v}{\partial \tau}(\tau, \gamma) = e_d(\gamma - \tau)$, we obtain the claimed formula:

$$e_d\eta - \lambda \frac{1 - F(\gamma)}{f(\gamma)}e_d + (1 + \lambda)(\gamma - \tau) = 0$$

□

A.3 Proof of Lemma 5

Proof. The first order condition from Lemma 4 yields:

$$\tau = \frac{\eta}{\lambda + 1} - \frac{\lambda}{\lambda + 1} \frac{1 - F(\gamma)}{f(\gamma)} + \gamma$$

Now using the uniform distribution assumption 2 yields:

$$\tau = \frac{\eta}{\lambda + 1} - \frac{\lambda}{\lambda + 1}(\bar{\gamma} - \gamma) + \gamma$$

Using the definition of μ , the first order condition becomes:

$$\tau = \mu\eta - (1 - \mu)(\bar{\gamma} - \gamma) + \gamma$$

Recall that the first order condition is only relevant for $\gamma \geq \gamma_L$. The types $\gamma < \gamma_L$ do not participate and thus choose $\tau = \gamma$. Thus the formula for τ holds.

Now recall that there exists γ_L such that all $\gamma < \gamma_L$ do not participate whilst all $\gamma > \gamma_L$ do participate. We can distinguish the following two cases:

case 1: $\gamma_L > \underline{\gamma}$

By the formula for the optimal $\tau(\gamma)$, this case corresponds to $\mu < \frac{\bar{\gamma} - \underline{\gamma}}{\eta + \bar{\gamma} - \underline{\gamma}}$ and in this case we have:

$$\gamma_L = \bar{\gamma} - \eta \frac{\mu}{1 - \mu}$$

The budget that needs to be spent is obtained by integrating the transfers, yielding:

$$m(\mu) := \frac{e_d \eta^3 (2 - \mu) \mu^3}{6(1 - \mu)^2 (\bar{\gamma} - \underline{\gamma})}$$

$m(\mu)$ is strictly increasing on $[0, 1]$, since $m'(\mu) = \frac{e_d \eta^3 \mu^2 (3 - (3 - \mu) \mu)}{3(1 - \mu)^3 (\bar{\gamma} - \underline{\gamma})}$.

Moreover:

$$m(0) = 0$$

$$m\left(\frac{\bar{\gamma} - \underline{\gamma}}{\eta + \bar{\gamma} - \underline{\gamma}}\right) = \frac{e_d \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}$$

This shows that the restriction of $\mu(M)$ to $[0, \frac{e_d \eta (\bar{\gamma} - \underline{\gamma})^2 (2\eta + \bar{\gamma} - \underline{\gamma})}{6(\eta + \bar{\gamma} - \underline{\gamma})^2}]$ is the unique inverse of the restriction of $m(\mu)$ to $[0, \frac{\bar{\gamma} - \underline{\gamma}}{\eta + \bar{\gamma} - \underline{\gamma}}]$.

case 2: $\gamma_L = \underline{\gamma}$

In this case, integrating the transfers yields:

$$\frac{e_d}{6} (\mu^2 (3\eta^2 + 3\eta(\bar{\gamma} - \underline{\gamma}) + (\bar{\gamma} - \underline{\gamma})^2) - (\bar{\gamma} - \underline{\gamma})^2)$$

This is strictly increasing in μ . □

A.4 Proof of Lemma 6

Proof. Consider first a reward payment scheme of the following form:

$$\tilde{t}(\tau) = \text{Max}(0, R(a\tau + b\tau^2 + c))$$

where R denotes the operator that preserves the right half (relative to its extremum) of the parabola $a\tau + b\tau^2 + c$ but outputs 0 on the left half.

We will show that there exist unique values for a, b, c as a function of $\mu(M)$ such that $\tilde{t}$ implements the optimal schedule of tax rates, which by Lemma 5 we know to be $\tau(\gamma) = \max(\gamma, \mu(M)\eta - (1 - \mu(M))(\bar{\gamma} - \gamma) + \gamma)$. We will then show that the corresponding α, β, τ^* are as claimed.

Consider a type γ and suppose that a, b, c are such that γ participates when faced with $\tilde{t}$, in the sense of optimally choosing a τ with $\tilde{t}(\tau) > 0$. The type τ has utility $v(\tau, \gamma) + \tilde{t}(\tau)$. By Lemma 2 we have $v(\tau, \gamma) = -\frac{1}{2}e_d(\tau - \gamma)^2$, so the first order condition for γ 's choice of τ becomes:

$$e_d(\tau - \gamma) = a + 2b\tau$$

Solving for τ gives:

$$\tau = \frac{a + e_d \gamma}{e_d + 2b} \quad (6)$$

By Lemma 5, we know that for the participating types, the optimal τ is given by:

$$\tau = \mu\eta - (1 - \mu)(\bar{\gamma} - \gamma) + \gamma \quad (7)$$

The functions defined by equations 6 and 7 are identical iff the following holds:

$$a = \frac{e_d \eta \mu - (1 - \mu) e_d \bar{\gamma}}{2 - \mu} \quad (8)$$

$$b = \frac{(1 - \mu)}{2(2 - \mu)} e_d \quad (9)$$

With this, one finds that if a type γ participates, then his optimal choice is $\mu\eta - (1 - \mu)(\bar{\gamma} - \gamma) + \gamma$, which coincides exactly with the choice at the optimal mechanism by Lemma 5.

Now we need to choose the constant c such that exactly the same types actually participate as at the optimal mechanism. Consider first the case with partial participation. By Lemma 5 we know that the minimal type that participates is $\gamma_L = \bar{\gamma} - \eta \frac{\mu}{1 - \mu}$ and this type is indifferent between participating or not. Moreover, this type chooses $\mu\eta - (1 - \mu)(\bar{\gamma} - \gamma_L) + \gamma_L = \gamma_L$. Thus under the quadratic reward payment scheme $\tilde{t}$, the type γ_L would get the following utility if it were to participate:

$$a\gamma_L + b\gamma_L^2 + c = a(\bar{\gamma} - \eta \frac{\mu}{1 - \mu}) + b(\bar{\gamma} - \eta \frac{\mu}{1 - \mu})^2 + c$$

Thus in order for γ_L to also be indifferent between participating or not, we need precisely the following:

$$c = -a(\bar{\gamma} - \eta \frac{\mu}{1 - \mu}) - b(\bar{\gamma} - \eta \frac{\mu}{1 - \mu})^2$$

Substituting in the expression 8 for a and expression 9 for b yields:

$$c = -\frac{e_d(\eta\mu + \bar{\gamma}(1 - \mu))^2}{2(2 - \mu)(1 - \mu)}$$

Now let us verify for the case with partial participation that $\tilde{t}$ is identical to the $t(\tau) = \alpha(\tau - \tau^*) + \beta(\max(0, \tau - \tau^*))^2$ as defined in the statement of the proposition. For $\tau \geq \tau^*$ we simply need to verify that the coefficients of the quadratic coincide, i.e. that the following holds:

$$a = \alpha - 2\beta\tau^*$$

$$b = \beta$$

$$c = -\alpha\tau^* + \beta\tau^{*2}$$

The computations verifying these identities are omitted here.

Since $\alpha = 0$, it follows that $\tilde{t}$ also agrees with the claimed t on the remaining part of the domain, i.e. where $\tau < \tau^*$.

Now consider the case with full participation. In this case we need to set c such that the type γ_L gets 0 utility. This is achieved by:

$$c = -\frac{e_d\gamma_L^2(1-\mu)}{2(2-\mu)}$$

Again, we can verify the identity of $\tilde{t}$ with the t defined in the proposition by comparing the coefficients. $\square$

A.5 Proof of Lemma 7

Proof. Consider a fixed M and let us vary η . We clearly must have $\lim_{\eta \rightarrow \infty} \mu = 0$, since μ is one over the marginal value of relaxing the budget constraint. Now the formula holding under partial participation, $\frac{e_d\eta^3(2-\mu)\mu^3}{6(1-\mu)^2(\bar{\gamma}-\underline{\gamma})} = M$, implies that $\lim_{\eta \rightarrow \infty} \eta\mu = (\frac{3(\bar{\gamma}-\underline{\gamma})}{e_d}M)^{\frac{1}{3}}$.

Hence the set of participating types converges to $[\bar{\gamma} - (\frac{3(\bar{\gamma}-\underline{\gamma})}{e_d}M)^{\frac{1}{3}}, \bar{\gamma}]$ and the tax rate converges to $\tau(\gamma) = \gamma + (\frac{3(\bar{\gamma}-\underline{\gamma})M}{e_d})^{\frac{1}{3}} - (\bar{\gamma} - \gamma)$. Moreover, by Lemma 6, the coefficients of the reward function become:

$$\alpha = 0, \beta = \frac{1}{4}e_d, \tau^* = \bar{\gamma} - (\frac{3(\bar{\gamma}-\underline{\gamma})}{e_d}M)^{\frac{1}{3}}$$

Now consider the case with full participation. We get:

$$\lim_{\eta \rightarrow \infty} \mu\eta = \sqrt{\frac{2M}{e_d}} + \frac{1}{3}(\bar{\gamma} - \underline{\gamma})^2$$

Hence

$$\lim_{\eta \rightarrow \infty} \tau(\gamma) = \lim_{\eta \rightarrow \infty} \mu\eta - (1-\mu)(\bar{\gamma} - \gamma) + \gamma = \gamma + \sqrt{2\frac{M}{e_d}} + \frac{1}{3}(\bar{\gamma} - \underline{\gamma})^2 - (\bar{\gamma} - \gamma)$$

The resulting emissions reductions are given by:

$$\int_{\gamma=\underline{\gamma}}^{\bar{\gamma}} e_d(\tau(\gamma) - \gamma)f(\gamma)d\gamma = e_d(\sqrt{2\frac{M}{e_d}} + \frac{1}{3}(\bar{\gamma} - \underline{\gamma})^2 - \frac{(\bar{\gamma}-\underline{\gamma})^2}{2})$$

Moreover, by Lemma 6, the coefficients of the reward function become:

$$\alpha = \frac{\sqrt{2\frac{M}{e_d} + \frac{1}{3}(\bar{\gamma}-\underline{\gamma})^2} - (\bar{\gamma}-\underline{\gamma})}{\sqrt{2}}e_d, \beta = \frac{1}{4}e_d, \tau^* = \bar{\gamma} - \sqrt{2}(\bar{\gamma}-\underline{\gamma}) + (1+\sqrt{2})\sqrt{2\frac{M}{e_d} + \frac{1}{3}(\bar{\gamma}-\underline{\gamma})^2}$$

$\square$

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